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RESEARCH ARTICLE

A Balanced Squad for Indian Premier League Using Modified NSGA-II

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ABSTRACT Selecting team players is a crucial and challenging task demanding a considerable amount of thinking and hard work by the selectors. The present study formulated the selection of an IPL squad as a multi-objective optimization problem with the objectives of maximizing the batting and bowling performance of the squad, in which a player's performance is estimated using an efficient Batting Performance Factor and Combined Bowling Rate. Also, the proposed model tries to formulate a balanced squad by constraining the number of pure batters, pure bowlers, and all-rounders. Bounds are also considered on star players to enhance the performance of the squad and also from the income prospects of IPL. The problem in itself is treated as a 0/1 knapsack problem for which two combinatorial optimization algorithms, namely, BNSGA-II and INSGA-II, are developed. These algorithms were compared with existing modified NSGA-II for IPL team selection and three other popular multi-objective optimization algorithms, NSGA-II, NSDE, and MOPSO-CD, on the basis of standard performance metrics: hypervolume, inverted generational distance, and number of Pareto optimal solutions. Both algorithms performed well, with BNSGA-II performing better than all the other algorithms considered in this study. The IPL 2020 players' data validated the applicability of the proposed model and algorithms. The trade-off squads contained players of each expertise in appropriate proportions. Further analysis of the trade-off squads demonstrated that many theoretically selected players performed well in IPL 2020 matches.

INDEX TERMS Cricket, twenty20, knapsack problem, combinatorial, Indian premier league, multi-objective, optimization, squad selection.

I. INTRODUCTION

The Indian premier league (IPL) is a franchise-based competition started in 2008 by the Board of control for cricket in India (BCCI) to promote cricket in India [1]. IPL is the most attended cricket league and has a significant contribution to the social and economic sectors. In 2015, IPL contributed ₹11.5 B to the Indian economy.

In this league, the players are bought by high-profile owners through an auction process. Each franchise owner spends a significant amount of money on purchasing players that have the ability to win and which also fit their business objectives. Considering the huge amount of money invested in the purchase of players, the franchise owners try to explore the best possible combinations of players for their squad.

In view of this, the researchers in [2] have formulated IPL team selection as a bi-objective optimization problem with the aim to maximize batting and bowling performances and satisfy the IPL regulations: budget constraint, bound on the

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number of wicket-keepers, captains, and overseas players. Similar objectives and constraints can be used to choose players for an IPL squad. However, since an all-rounder contributes to both batting and bowling performances while a pure batter (pure bowler) primarily contributes to squad batting performance (bowling performance), the trade-off squads for this problem, obtained using an optimization algorithm mainly contain all-rounders, which is a feasible solution theoretically but not a practical one. Keeping this drawback in mind, a new model is proposed in the present study in order to choose a squad for the IPL which is more balanced. The proposed model has additional bounds on the number of players based on their expertise, ensuring a sufficient proportion of pure batters, pure bowlers and all-rounders in a squad.

Further, only a player's batting average (batting strike rate) was used in the existing multi-objective team selection models in [2] and [3] to determine its batting performance (bowling performance). However, in IPL, maintaining a high strike rate is the primary focus of a batter because, in a short cricket format, scoring runs rapidly is more important for a team's success than the average run per innings. On the other hand, the efforts of bowlers with a good economy rate (bowlers who prevent the batters from scoring runs) and the pressure to maintain a high strike rate increase the chances of batters being dismissed. Therefore, we use more efficient measures to better estimate a player's performance: the batting performance factor [4] and the combined bowling rate [5]. These measures also include the batting strike rate (bowling strike rate and bowling economy) along with the batting average (bowling average) for computing the batting (bowling performance) of a batter (bowler).

A trade-off squad may contain many players who have participated only in a few matches because these players have strong statistics and are less expensive than more experienced players. Instead, each squad should have some star players who not only contribute to the success of the franchise through their performance but also through their popularity and fame. Therefore, we put bounds on the minimum number of star batters, star bowlers, and star all-rounders so that the squad contains a sufficient number of star players within the same budget. Additionally, we use data from all the franchise leagues and T20 international matches to more accurately estimate a player's performance.

The Knapsack problems are among the most studied combinatorial optimization problems (COPs), finding solutions to which remain an exciting challenge [6]. The proposed IPL squad selection problem can be viewed as a knapsack problem with the objective to choose limited players from available such that the total performance of the squad is maximized under the fixed budget. In IPL 2020, there are 10^{27} to 10^{30} possible combinations for selecting a squad of 21 to 25 players out of 185 auctioned players. Such a large number of combinations makes the selection of players a challenging COP.

Various optimization methods are available in the literature for cricket team selection: genetic algorithm for selecting an optimal national cricket team [7] and integer programming for selecting playing XI team [8], T20 world cup team [9], and IPL team [3]. The multi-objective IPL team selection model has been solved using integer linear programming [3] and non-dominated sorting genetic algorithm-II (NSGA-II) [2], [10]. NSGA-II [11] is a population-based multi-objective optimization algorithm widely used to solve the multi-objective COPs [12], including multi-objective knapsack problems [13], [14].

The present study developed two NSGA-II variants, BNSGA-II and INSGA-II, for solving the proposed multi-objective IPL squad selection problem. BNSGA-II has a binary chromosome representation, and INSGA-II has an integer chromosome representation. The problem constraints are handled using the constraint dominance principle, and a repair mechanism is proposed to repair the infeasible solutions that occur in BNSGA-II due to the violation of a single constraint in order to produce a sufficient number of feasible solutions. The efficiency of the proposed algorithms is validated by comparing them with NSGA-II, an algorithm for the bi-objective IPL team selection problem [2] and three other popular multi-objective optimization algorithms: NSGA-II [11], non-dominated sorting differential evolution (NSDE) [15], and multi-objective particle swarm optimization with crowding distance (MOPSO-CD) [16], using hypervolume, inverted generational distance (IGD), number of Pareto optimal solutions (NPS), and computational time. Next, the trade-off squads obtained through the best-performed algorithm are analyzed based on their cost and the fielding performance. The performance of trade-off squad players in IPL 2020 validates the efficiency of the proposed model. A practical situation is also simulated where a franchise owner favours certain players irrespective of their cost.

In summary, the main contribution of this study is:

- Using the concept of knapsack problem for selecting an IPL team.
- Designing a new model for selecting a balanced squad for IPL.
- Developing two new algorithms, BNSGA-II and INSGA-II, for solving the proposed model.

The rest of this paper is organized as follows: Section 2 describes the related work. Section 3 defines the multi-objective knapsack problem. Section 4 formulates the proposed IPL Squad selection problem. Section 5 describes the proposed methodology. Section 6 discusses the experiment results and comparison. Section 7 shows the analysis and discussion. Finally, the conclusion is presented in Section 8.

II. RELATED WORK

This section discusses the optimization methods available in the literature to select IPL team/squad players. In addition to the optimization methods, researchers also used methods based on other techniques, such as data envelopment analysis [17], [18] and machine learning algorithms [19], [20].

However, this study is only focused on optimization methods for single and multi-objective optimization models. The comparative study between the related work on optimization methods for cricket team selection and the proposed work is given in Table 1.

Sathya and Jamal [7] proposed genetic algorithm for selecting eleven out of fifty national players for a one-day international(ODI). The factors such as the number of pacers and spinners, composition of left-hander and right-hander, and partnership records to select a more flexible, balanced, and diverse team had been considered for selection purposes.

Ahmed *et al.* [2] proposed NSGA-II with a novel gene representation and decision-making techniques for IPL T20 team selection. The resulting teams were compared with IPL 4th edition teams and found better theoretically. The authors also demonstrated a dynamic auction-based player selection to make the procedure more realistic.

Bhattacharjee and Saikia [9] selected an optimal squad of 15 players using binary integer programming and compared it with the Indian team selected by the International cricket council (ICC) for the corresponding T20 world cups. Again in 2016, authors [8] used binary integer programming to select the playing XI team, which took care of the area of specialization of the captain and other players.

Chand *et al.* [3] proposed an integer programming method for IPL team selection that guaranteed optimality and demonstrated its scalability using two-objective, three-objective, and five-objective formulations. The objectives of the problem were based on batting, bowling, fielding, cost, and star power. The construction of partial teams was done around the preferred players. In addition, the ranking of players was done using the players' performance to aid the decision-making.

III. MULTI-OBJECTIVE KNAPSACK PROBLEM

Given a knapsack with weight capacity W and a set of n items in which i^{th} item has weight $w_i \leq W$ and profit per objective $v_i^k > 0$, $i = 1, 2, \ldots, n$, then the multi-objective 0/1 knapsack problem aims to fill the knapsack with given items within its capacity such that the total profit is maximized. Mathematically, a K-objective knapsack problem is defined as follows:

$$Maximize \left(\sum_{i=1}^{n} v_i^1 x_i, \sum_{i=1}^{n} v_i^2 x_i, \ldots, \sum_{i=1}^{n} v_i^K x_i \right)$$
$$s.t \quad \sum_{i=1}^{n} w_i x_i \leq W$$

where

$$x_i = \begin{cases} 1, & item\ i\ is\ selected \\ 0, & item\ i\ is\ not\ selected \end{cases}, i = 1, 2, \ldots, n$$

IV. IPL SQUAD SELECTION PROBLEM

The IPL squad selection problem is similar to the above-defined multi-objective 0/1 knapsack problem, as illustrated in Table 2. The number of players to be selected for an IPL squad is limited; a franchise owner will either select a player (Yes =1) or not (No=0). Therefore, the concept of the 0/1 knapsack problem is used in this paper for selecting a player for an IPL squad.

A. PROBLEM DEFINITION AND FORMULATION

A balanced squad contains players of each expertise in suitable proportions. The proposed squad selection problem aims to form a balanced IPL squad with maximum net batting and bowling performance. The objectives and constraints of the problem are detailed in the following subsections. The variables and parameters used in the paper are defined in Table 3.

1) BATTING PERFORMANCE

The batting average[1] indicates the run-scoring capability of a player and estimates its batting performance. However, for limited-overs cricket, slow batting will lead to defeat rather than victory [4]; therefore, players with high batting strike rate[2] are also needed since it indicates the rate of scoring runs by a player.

It is not necessary that the players with high batting averages also have high strike rates; for example, the IPL 2020 auctioned players with high strike rates and low batting averages are shown in Table 4. The relation between these batting statistics can further be visualized using Fig.1. Here, the value of these statistics for pure batters and batting all-rounders are normalized (due to much difference in their scale) and sorted in decreasing order of the batting averages. The figure shows that the batting averages of the players decrease continuously, but their strike rates fluctuate. Also, they do not have any relation such that considering one will count both of them. Therefore, both factors should be considered important for analyzing a player's batting performance.

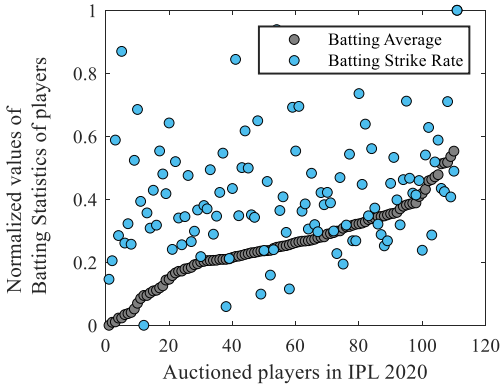


FIGURE 1. Relation between batting statistics of auctioned players in IPL 2020.

Researchers in [4] have proposed an efficient Batting performance factor (BPF) for calculating the player's batting performance. BPF is the weighted product of the normalized values of batting strike rates and batting averages, as given in (1). The higher value of α indicates a higher weightage to

[1]Batting average = Total runs scored/Number of times out.
[2]Batting strike rate = 100 × (Total runs scored/Total balls faced).

TABLE 1. Comparison of the related work and the proposed work.

Year	Goal	Constraints	Batting & Bowling Statistics	Method	Format	Data	Number of objectives	Ref.
2009	Selection of 11 players out of 50 national players	-	ICC rankings, statistics & cricket experts.	Genetic Algorithm	ODI	NA	Single	[7]
2013	IPL team selection	Captain, wicket-keeper, overseas player and budget	Batting, bowling, & fielding averages	NSGA-II + knee region approach	IPL	IPL 4th edition	Multi	[2]
2014	T20 world cup team selection	Openers, Middle-order batters, spin bowlers, fast bowlers, all-rounders, wicket-keeper	Batman's average & strike rate; Bowler's economy, average & strike rate	Binary Integer Programming	IPL	IPL 2009, 2010 & 2012	Single	[9]
2016	Selection of playing XI team	Openers, Middle-order batters, spin bowlers, fast bowlers, all-rounders, wicket-keeper	Batman's average & strike rate; Bowler's economy, average & strike rate	Binary Integer Programming	IPL	IPL 5th edition	Single	[8]
2018	IPL team selection	Captain, wicket-keeper, overseas player, budget, bowling average, batting average, fielding average, star-power	Batting, bowling, & fielding averages	Integer Linear Programming	IPL	IPL 4th edition	Multi /Many	[3]
Proposed work	Selection of balanced squad for IPL (23 out of 185 players)	Captain, wicket-keeper, overseas player, budget, pure batters, pure bowlers, all-rounders, star batters, star bowlers, star all-rounders	Batting Performance Factor [4] (Batman's average & strike rate); Combined Bowling rate [5] (Bowler's economy, average & strike rate)	Modified NSGA-II: BNSGA-II and INSGA-II	IPL	Franchise leagues & T20 international matches data up to August 2020	Multi	-

TABLE 2. Similarity between knapsack problem and IPL squad selection problem.

Knapsack problem	IPL squad selection problem
Knapsack	Squad
Item	Player
Item's weight	Auctioned price of the player
Item's profit	Performance of the player
Knapsack capacity	Maximum salary cap

batting strike rate than the batting average. Since this factor utilizes both batting statistics and one statistic can be prioritized over the other, we have used this factor to calculate the batting performance. The higher value of BPF indicates the higher batting performance of a player.

$$BPF = Batting\ strike\ rate^{\alpha} \times Batting\ average^{1-\alpha} \quad (1)$$

where $0 \leq \alpha \leq 1$

The first objective of this study is to maximize the net batting performance (or minimize the –net batting performance) of the squad as given in (2). Here, the objective function f_1 denotes the negative of the sum of BPF factors of selected players.

$$Minimize\ f_1 = -\sum_{i=1}^{n} x\,(i) \times BPF\,(i) \quad (2)$$

TABLE 3. Notations.

Notations	Definitions
m	Total number of players
N	Number of players in a squad
T	Maximum salary cap
a_1	Lower bound for the sum of the number of pure batters and batting all-rounders in a squad
a_2	Lower bound for the sum of the number of pure bowlers and bowling all-rounders in a squad
m_1	Lower bound for the number of pure batters
m_2	Lower bound for the number of pure bowlers
m_3	Upper bound for the number of all-rounders
n_1	Lower bound for the number of star pure batters
n_2	Lower bound for the number of star pure bowlers
n_3	Lower bound for the number of star all-rounders
BPF	Batting performance factor
CBR	Combined bowling rate
α	Weight ∈ (0,1)
P_m	Mutation probability
P_c	Crossover probability

where

$$x\,(i) = \begin{cases} 1, & if\ player\ i\ is\ chosen\ for\ the\ squad \\ 0, & otherwise \end{cases}$$

TABLE 4. IPL 2020 auctioned players with a high batting strike rate compared to batting average.

Players	Strike rate (Max Value = 175.76)	Batting average (Max value = 64.44)
Andre Russell	171.48	26.72
Krishnappa Gowtham	166.67	15.47
Fabian Allen	164.90	25.25
Harshal Patel	153.82	17.81
Sunil Narine	147.05	14.84

Here, n is the total number of players, and $BPF(i)$ is the batting performance factor of the ith player, $i = 1, 2, \ldots, n$.

2) BOWLING PERFORMANCE

The bowling average,[3] bowling strike rate,[4] and bowling economy[5] are the three factors that analyze a bowler's performance. In T20, due to a limited number of overs, pure batters usually need to risk their wicket to score more runs because of which bowlers with an excellent economy rate (i.e., bowlers that do not provide runs to batters) produce wickets for other bowlers.

The relationship between bowling statistics is analyzed by normalizing and sorting them according to bowling averages, as shown in Fig. 2. The figure illustrates that a bowler with a relatively higher bowling average does not necessarily have a relatively higher economy rate and bowling strike rate.

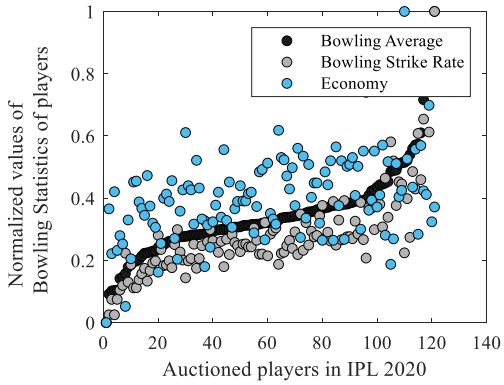


FIGURE 2. Relation between bowling statistics of auctioned players in IPL 2020.

Lemmer [5] has proposed Combined bowling rate (CBR), a potential formula for calculating the bowling performance of a player. CBR counts all three bowling statistics (3); therefore, we have used it to estimate players' bowling performance. The lower CBR indicates the higher bowling performance of a player.

$$CBR = \frac{1}{\frac{1}{Bowling\ average} + \frac{1}{Bowling\ economyrate} + \frac{1}{Bowling\ strike\ rate}} \tag{3}$$

[3]Bowling average = Total runs conceded /Total wickets taken.
[4]Bowling strike rate = Total balls bowled/Total wickets taken.
[5]Bowling economy rate = 6 × (Total runs conceded/Total overs bowled).

The second objective of this study is to maximize the net bowling performance (or minimize the -net bowling performance) of the squad as given in (4). Here, the objective function f_2 denotes the sum of the CBRs of selected players.

$$Minimize\ f_2 = \sum_{i=1}^{n} x\,(i) \times CBR\,(i) \tag{4}$$

$$where\ x\,(i) = \begin{cases} 1, if\ player\ i\ is\ chosen\ for\ the\ squad \\ 0, otherwise \end{cases}$$

Here, n is the total number of players, and *CBR(i)* is the combined bowling rate of the ith player , $i = 1, 2, \ldots, n$.

Note: Since a lower *CBR* is favourable, batters will be selected as a bowler due to their lower *CBR* and not because of their excellent batting statistics. Therefore, the *CBR* of a batter is taken 100 as a penalty.

3) CONSTRAINTS BASED ON IPL REGULATIONS

BCCI puts some constraints (5-9) on franchises for selecting players of an IPL squad. Constraint (5) limits the salary cap of franchises for purchasing players. Constraints (6) and (7) bounds the total number of players and the number of overseas players in a squad, respectively. Here m is the number of players in a squad, and T is the maximum salary cap. The other regulations (8-9) include the restrictions on the number of captains and wicket-keepers in the squad.

$$Total\ cost\ of\ squad \leq T \tag{5}$$
$$Number\ of\ players\ in\ a\ squad = m \tag{6}$$
$$Number\ of\ overseas\ players \leq 8 \tag{7}$$
$$Number\ of\ captains \geq 1 \tag{8}$$
$$Number\ of\ wicketkeepers \geq 1 \tag{9}$$

4) BOUND ON THE SUM OF THE NUMBER OF PURE BATTERS (BOWLERS) AND BATTING (BOWLING) ALL-ROUNDERS

The trade-off squads created by applying the above IPL regulations may contain mostly all-rounders as they improve both the batting and bowling performance of a squad. Also, due to budget limitation, bowlers, bowling all-rounders, and new players have a higher chance of being selected as these players have lower auction prices compared to pure batters and experienced players. Therefore, two additional constraints are considered, one bounding the sum of the number of pure batters and batting all-rounders (10) and the other bounding the sum of the number of bowlers and bowling all-rounders (11).

$$Sum\ of\ number\ of\ batters\ \&\ batting\ all-rounders \geq a_1 \tag{10}$$

$$Sum\ of\ number\ of\ bowlers\ \&\ bowling\ all-rounders \geq a_2 \tag{11}$$

Here, a_1 is the lower bound for the sum of the number of batters and batting all-rounders and a_2 is the lower bound for the sum of the number of bowlers and bowling all-rounders in a squad.

5) BOUND ON THE NUMBER OF PURE BATTERS, PURE BOWLERS, ALL-ROUNDERS, STAR BATTERS, STAR BOWLERS, AND STAR ALL-ROUNDERS

As the pure batters have higher *CBR* (as a penalty) than batting all-rounders, the trade-off squads still have the possibility of a shortage of batters. Therefore, instead of constraints (10-11), bounds are considered on the minimum number of pure batters (12), pure bowlers (13) and all-rounders (14).

$$Number\ of\ pure\ batters \geq m_1 \tag{12}$$
$$Number\ of\ pure\ bowlers \geq m_2 \tag{13}$$
$$Number\ of\ all-rounders \leq m_3 \tag{14}$$

where m_1 is the lower bound for the number of pure batters, m_2 is the lower bound for the number of bowlers, and m_3 is the upper bound for the number of all-rounders in a squad.

The above three constraints can be used to obtain a balanced squad. However, every franchise wants to have a sufficient number of star players[6] because a star player increases the squad's performance and draws public attention owing to its outstanding performance. Consequently, we have considered the bounds on the number of star players also (15-17).

$$Number\ of\ star\ pure\ batters \geq n_1 \tag{15}$$
$$Number\ of\ star\ pure\ bowlers \geq n_2 \tag{16}$$
$$Number\ of\ star\ all-rounders \geq n_3 \tag{17}$$

where n_1, n_2 and n_3 are respectively the lower bounds for the number of star pure batters, star pure bowlers, and star all-rounders in a squad.

B. DATASET

The proposed model is validated using the performances of IPL 2020 auctioned players in T20 tournaments, franchise leagues, and international matches. The data of 185 players up to August 2020 is taken from the website http://bigbashboard.com (a well-known web forum containing data on all cricket matches).

The details of the players are given in Table 5. The star players are subjectively considered based on the players' records in T20I and IPL. A star batter is a player who has either scored $\geq$ 1,000 runs in T20I (i.e., T20 International) or has scored $\geq$ 2,000 runs in IPL. Likewise, a star bowler is a player who has taken either $\geq$ 30 wickets in T20I or $\geq$ 60 wickets in IPL. Further, for being a star all-rounder, a player should have scored $\geq$ 300 runs and taken $\geq$ 15 wickets in T20I or $\geq$ 600 runs and taken $\geq$ 30 wickets in IPL. The total number of star players is 70, out of which 24 are pure batters, 26 are pure bowlers, and 20 are all-rounders.

C. PROBLEM COMPLEXITY

The complexity of the proposed COP can be defined as the possible number of players combinations for a squad, *i.e.*, c_m^n, where n is the total number of selected players and m is the maximum number of players in a squad. The complexity varies with the number of players in the squad. For, $m = 23$ and $n = 185$, it is $\sim 10^{29}$.

[6]Player with high-performance records in IPL and T20I is considered a 'star player'.

TABLE 5. Details of Auctioned players in IPL 2020.

Role	Number of players
Pure Batter	64
Pure Bowler	74
Wicket-keeper	24
All-rounder	47
Batting all-rounder	34
Bowling all-rounder	33
Captain	23

V. PROPOSED ALGORITHMS

NSGA-II, introduced by Deb *et al.* [11], is a popular optimization algorithm capable of solving difficult multi-objective optimization problems. The present study develops two NSGA-II variants to solve the proposed IPL squad selection problem. The following are the basic operations of the proposed algorithms:

A. CHROMOSOME REPRESENTATION

The performance and functionality of an algorithm are affected by a solution representation. Therefore, two efficient solution representations, binary and integer chromosome representations, are defined to solve the proposed knapsack problem, as shown in Figures 3 and 4. Both of these representations denote a squad with m number of players. In Fig. 3, the player tag value corresponding to gene value '1' represents the index of the player selected for the squad. This representation maintains a unique combination of genes for each squad; no operations cause the re-occurrence of players in a squad. In Fig. 4, the gene values corresponding to player tags represent the indices of the selected players.

Player Tag	1	2	3	4	…	n
Chromosome	1	0	0	1	…	0

FIGURE 3. Binary chromosome representation.

Player Tag	1	2	3	4	…	m
Chromosome	15	14	6	21	…	3

FIGURE 4. Integer chromosome representation.

The algorithm with binary chromosome representation is named 'BNSGA-II', and the algorithm with integer chromosome representation is named 'INSGA-II.'

B. POPULATION INITIALIZATION

In BNSGA-II, the initial population is randomly generated as binary chromosomes with m number of 1s. The initial population in INSGA-II is also randomly generated as integer chromosomes having m unique values out of n.

C. CROSSOVER AND MUTATION OPERATORS

In BNSGA-II, the two-point crossover and swap mutation are used, as shown in Fig. 5. The two-point crossover exchanges the genes between two random crossover points k_1 and k_2 of the parent chromosomes to generate the offspring chromosomes, where k_1 and k_2 are random integers in $[1, \lfloor n/2 \rfloor]$ and $[k_1 + 1, n]$, respectively. Similarly, the swap mutation exchanges two random genes of the offspring population.

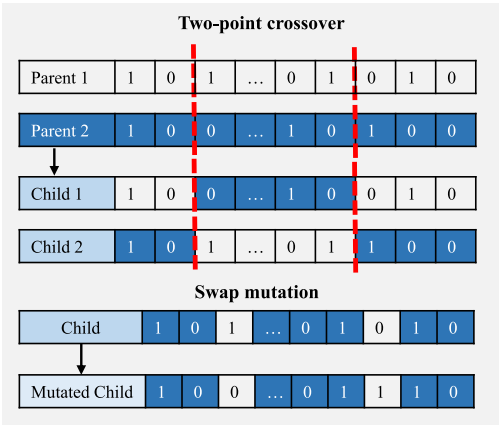


FIGURE 5. BNSGA-II crossover and mutation operators.

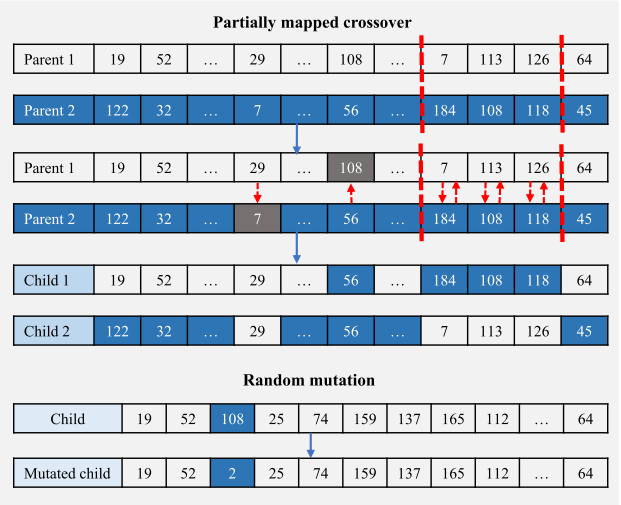


FIGURE 6. INSGA-II crossover and mutation operators.

In INSGA-II, partially mapped crossover [21] and random mutation are used, as shown in Fig. 6. This crossover exchanges a part of the parent chromosome with an operation involving a mapping procedure to maintain the uniqueness of players in the squad. Further, random mutation is used in which a random gene (a random player) is replaced with a gene different from the chromosome genes (player not present in the squad).

D. CONSTRAINT HANDLING

The constraint dominance principle [11] is used to take care of the constraints. According to this principle, if there are two solutions x^1 and x^2, then the selection is made as per the following three possibilities:

1) If both x^1 and x^2 are feasible, where one that dominates the other is selected.
2) If one solution is infeasible, then the feasible solution is considered.
3) If both x^1 and x^2 are infeasible, then the one with a lesser constraint violation is said to dominate the other and is selected.

E. REPAIR MECHANISM

In BNSGA-II, initially, each squad has m number of players, but after the first generation, the constraint (6) of maintaining the fixed number of players in each squad starts violating. Due to this constraint, the infeasibility in the population results in fewer feasible population members despite taking a large population size. Therefore, a repair mechanism is proposed to repair the offspring population in each generation to produce sufficient feasible population members for the next generation. More clearly, the repair mechanism helps remove the large infeasibility due to a single constraint. The pseudocode for the repair mechanism is given in Algorithm 1.

Algorithm 1 Repair Mechanism

Parameters: *Mutated chromosome (x), Number of variables (n), Squad length (m)*
1: Find the number of non-zero elements in x and their indices set: ***N and Index***
2: IF N> m, then
  x(Index)=0
  Randomly select m indices from the set Index: ***Index1***
  x(Index1)=1
ELSE IF N< m, then
  Take the compliment of the set Index: ***Index2***
  Randomly select m-N indices from Index2: ***Index3***
  x(Index3)=1
END IF

A brief summary of BNSGA-II and INSGA-II operators is provided in Table 6. The other procedures, including the non-dominating sorting, crowding distance calculation, and selection procedures, are the same as in conventional NSGA-II [11]. The algorithmic steps of the BNSGA-II are shown in Algorithm 2.

VI. EXPERIMENTAL RESULTS AND COMPARISONS

This section designs experiments considering two cases for IPL squad selection. Case-I is an optimization problem with objectives (2) and (4) subject to constraints (5-11), and case-II is the optimization problem with objectives (2) and (4) subject to constraints (5-9) and (12-17). The first experiment

TABLE 6. Proposed algorithms.

Operators	BNSGA-II	INSGA-II
Chromosome representation	Binary	Integer
Crossover	Two-point crossover	Partially mapped crossover
Mutation	Swap mutation	Random mutation
Constraint handling approach	Constraint dominance principle + Repair mechanism	Constraint dominance principle

validates the efficiency of the proposed algorithms and the second experiment compares the cases for a balanced squad. The setup for the experiments and the results are detailed in the following subsections.

A. EXPERIMENTAL SETUP

This section introduces the other algorithms used for comparison, parameter settings, and performance metrics.

1) ALGORITHMS FOR COMPARISON

The efficiency of BNSGA-II and INSGA-II is validated using four prominent algorithms. One of them is the modified NSGA-II for the bi-objective IPL team selection problem (naming convention used in this study: 'MNSGA-II') [2], which used a chromosome representation with fixed positions of the captain and wicket-keeper. MNSGA-II was originally coded for an 11 players team, and the current study extends it to a squad of 23 players.

Player Tag	1	2	3	4	…	n
Chromosome	0.11	0.53	0.36	0.12	…	0.66
Rounding	1	0	0	1	…	0

FIGURE 7. Real chromosome representation.

The other popular multi-objective optimization algorithms that are used for comparison with the proposed algorithms are real-coded NSGA-II (naming convention used in this study: 'RNSGA-II') [11], NSDE [15], and MOPSO-CD [16]. The chromosome representation used for these three algorithms is given in Fig. 7. Here, the continuous representation is mapped to binary representation with the help of a rounding function. Suppose $(x_1, x_2, \ldots, x_n)$ is the chromosome with continuous representation; then it gets round to 0-1 representation $(b_1, b_2, \ldots, b_n)$ using (18). The rounding function selects m players (or nearly m) out of n players to satisfy the constraint (6).

$$b_i = \begin{cases} 1, & x_i \leq (m/n) \\ 0, & otherwise \end{cases}, i = 1, 2, \ldots, n \tag{18}$$

Algorithm 1 repairs the infeasible chromosomes obtained using these three algorithms.

2) PARAMETER SETTING

The parameters for the proposed model and algorithms are given in Table 7. The population size and the number of generations are taken after fine-tuning the algorithm. Several combinations were tried to optimize the performance of the algorithm. It was observed that the population size of 200 and the stopping criteria of 500 generations gave the best possible results. Likewise, the other parameters were also fine-tuned so that the Pareto optimal solutions cover the whole Pareto front and with good convergence. Considering the brevity of space only, the best parameters are shown in the study.

Algorithm 2 BNSGA-II

Algorithm parameters: *Population size (pop_size), Number of variables (n), Maximum number of generations (max_gen), Mutation probability (p_m), Crossover probability (p_c)*
Problem parameters: *chromosome (x), Squad length (m)*
//Population initialization
1: FOR i = 1: pop_size
1.1: Generate a matrix $[A]_{1\times m}$ with unique elements $a_i \in \{1, 2, \ldots, n\}$
1.2: FOR j=1: n
IF $j \in A$ then
x (i, j) =1
ELSE
x (i, j) = 0
END IF
END FOR
END FOR
// Generating next generations P_t, t=2, 3, …, max_gen
2: FOR t=1: max_gen
2.1: Perform two-point crossover on P_t using p_c
2.2: Perform swap mutation and generate the offspring population using
2.3: Perform the repair mechanism on the offspring populationusing Algorithm 1
2.4: Combine parent population and repaired offspring population
2.5: Put the combined population into different non-dominated fronts $\mathcal{F} = (\mathcal{F}_1, \mathcal{F}_2, \ldots,)$ using non-dominated sorting procedure
2.6: Set the parent population $P_{t+1}=\varphi$, k=1
2.7: WHILE ($P_{t+1} \leq$ pop_size) DO
IF ($P_{t+1} \cup \mathcal{F}_k \leq$ pop_size) then
Include the kth non-dominated front $\mathcal{F}_k$ in the P_{t+1}(P_{t+1}= $P_{t+1} \cup \mathcal{F}_k$)
ELSEIF ($P_{t+1} \cup \mathcal{F}_k >$ pop_size) then
Calculate the crowding distance of members of the front $\mathcal{F}_k$
Sort the members of $\mathcal{F}_k$ in decreasing order of crowding distance
Add the top (pop_size $- |P_{t+1}|$) members of $\mathcal{F}_k$ to P_{t+1}
END IF
END WHILE
END FOR
*3:**Output**: A set of Pareto optimal solutions $\mathcal{F}_k$*

The model parameters $a_1, a_2, m_1, m_2, m_3, n_1, n_2, n_3$ are fixed for the squad length $m = 23$, as shown in Table 7 and can be changed with the value of m. The value of parameter α is set using sensitivity analysis as follows:

SENSITIVITY ANALYSIS OF PARAMETER α: The parameter α calculates the batting performance of the squad by prioritizing one batting statistic over the other; therefore, a sensitivity analysis is performed to decide its value. For

TABLE 7. Parameter settings for algorithms and problem model.

Algorithms parameters							Model parameters	
Algorithms	Population Size	Maximum Generations	Crossover Probability	Mutation Probability	Mutation Rate	Other Parameters	Parameter	Value
MOPSO-CD	200	500	-	-	0.5	Inertia rate = 0.4	α	0.6
NSDE			-	-	0.5	Scaling factor = 0.2	m	23
RNSGA-II			1	0.1	-	-	a_1 , a_2	11
BNSGA-II			1	0.1	-		m_1, m_2	8
INSGA-II			1	0.1	-		m_3	6
MNSGA-II			0.9	0.05	-	Crossover rate = 0.1 SBX distribution index=10, mutation distribution index = 20	n_1, n_2, n_3	3
							T	₹ 8500 L

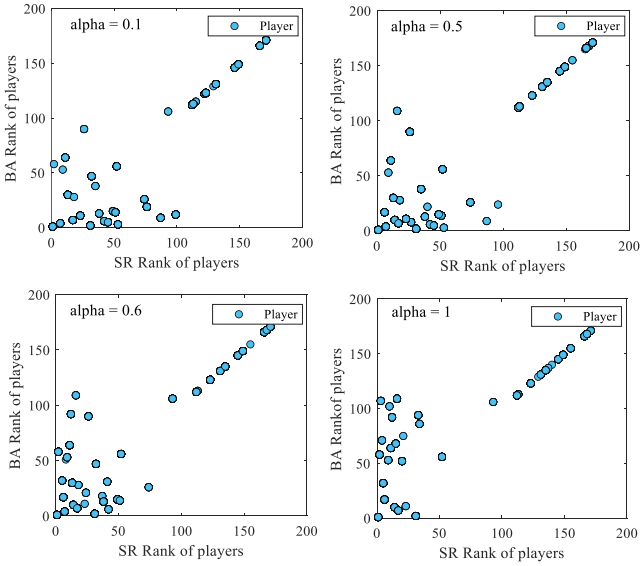


FIGURE 8. Sensitivity analysis of alpha parameter.

this purpose, we first obtain the trade-off squads for different values of α. Then, the players who occur in these squads are given 'BA Rank' ('SR Rank') according to their batting averages (batting strike rates) such that the higher the batting average (batting strike rate) of a player, the better will be its BA Rank (SR rank). The player with the highest batting average (batting strike rate) has 1st BA rank (1st SR rank). The relation between BA rank and SR rank for $\alpha = 0.1$, $\alpha = 0.5$, $\alpha = 0.6$, and $\alpha = 1$ is shown in Fig. 8. For lower α, there is not much difference in both ranks of players, but this difference increases as the value of α increase from 0.5 to 1. As the T20 is a shorter format than ODI, more hitters, i.e., batters with superb strike rates, are needed to achieve a good score. Therefore, the value of α is set to 0.6 for the proposed model.

3) PERFORMANCE METRICS

The performance metrics NPS, hypervolume, and IGD are used for comparing the algorithms [22]. These are the most popular metrics for comparing the multi-objective optimization algorithms and are defined as:

a: NPS

Let S_i be the ith solution set, and S_{NDi} are the non-dominated solutions in S_i, then the NPS in S_i is $|S_{NDi}|$. This measure compares the NPS of different solution sets.

b: HYPERVOLUME

This is the volume of the region dominated by the solution set S in the objective space. Mathematically, $\forall x_i \in S$, a hypercube v_i is constructed with a reference point W and solution x_i as diagonal corners of the hypercube, the reference point W can be found by constructing a vector of worst objective values. After that, a union of all hypervolumes is calculated as:

$$HV = volume(\bigcup_{i=1}^{|S|} v_i) \tag{19}$$

c: IGD

This measure evaluates the proximity of the solution set S to the solution set S^* and is defined as the average distance of each reference solution $y \in S^*$ from its nearest solution in S as:

$$IGD = \frac{1}{|S^*|} \sum_{y \in S^*} min\{d_{xy}/x \in S\} \tag{20}$$

where, d_{xy} is the Euclidean distance between solution $x \in S$ and $y \in S^*$.

4) COMPUTATIONAL DETAILS

The algorithms are coded in MATLAB and executed on Intel (R) Core (TM) i7-3770 with a 3.40 GHz processor with 12 GB of RAM.

B. EXPERIMENTAL RESULTS

This section compares the solutions obtained using various algorithms for the proposed problem. The simulation is also performed by fixing the preferred players and building partial squads around them.

1) PERFORMANCE COMPARISON OF ALGORITHMS

An experiment is designed in which six algorithms, MNSGA-II, NSDE, RNSGA-II, MOPSO-CD, INSGA-II, and BNSGA-II, are compared using three performance metrics (hypervolume, NPS, IGD), and computational time.

TABLE 8. Comparison of algorithms' performance for case-I and case-II.

Case-I	NPS	Time (in seconds)		Hypervolume				IGD			
Algorithm	Avg. value	Avg. value	Std. Dev	Max. Value	Avg. value	Median. value	Std. Dev.	Min. Value	Avg. value	Median. value	Std. Dev.
MNSGA-II	58.7250	109.6087	5.1356	0.8716	0.8178	0.8196	0.0255	0.0323	0.0712	0.0675	0.0210
MOPSO-CD	34.8500	388.9935	30.8479	0.8903	0.8427	0.8460	0.0322	0.0260	0.0509	0.0472	0.0233
NSDE	159.0250	495.6788	44.1512	0.9382	0.9279	0.9278	0.0053	0.0050	0.0115	0.0109	0.0034
RNSGA-II	123.9500	515.8244	32.9480	0.9335	0.9202	0.9218	0.0082	0.0062	0.0115	0.0110	0.0028
INSGA-II	164.3000	**108.9525**	8.6757	0.9381	0.9266	0.9263	0.0063	0.0052	0.0125	0.0117	0.0046
BNSGA-II	**165.0250**	129.0056	13.5195	**0.9504**	**0.9442**	**0.9454**	0.0044	**0.0015**	**0.0038**	**0.0032**	0.0022
Case-II	NPS	Time (in seconds)		Hypervolume				IGD			
Algorithm	Avg. value	Avg. value	Std. Dev	Max. Value	Avg. value	Median. value	Std. Dev.	Min. Value	Avg. value	Median. value	Std. Dev.
MNSGA-II	27.4250	**98.5621**	6.8861	0.6191	0.5770	0.5805	0.0283	0.0759	0.111	0.104	0.0253
MOPSO-CD	20.5750	387.9676	62.3816	0.6084	0.5610	0.5657	0.0262	0.0846	0.1220	0.1189	0.0242
NSDE	31.5500	283.9193	45.1264	0.6983	0.6779	0.6779	0.0111	0.0216	0.0360	0.0365	0.0065
RNSGA-II	32.5000	480.0477	31.9278	0.7027	0.6789	0.6793	0.0125	0.0239	0.0358	0.0350	0.0067
INSGA-II	51.3000	114.5924	10.7036	0.7222	0.7056	0.7066	0.0095	0.0120	0.0200	0.0191	0.0041
BNSGA-II	**57.2500**	111.0790	8.3994	**0.7294**	**0.7167**	**0.7185**	0.0090	**0.0110**	**0.0162**	**0.0156**	**0.0031**

Before this experiment, for both cases, all the algorithms were run independently 40 times such that each run started from different populations to avoid biased and unfair comparisons.

For experiment purposes, we obtain a Pareto optimal set (POS) for each run of the algorithms. Every POS is normalized by normalizing each kth objective function value f_k using the relation (21), where F is the combination of POS of the six algorithms ($6 \times 40 = 240$POSs). The normalized objective function values f_k' lies in the interval [0,1]; therefore, (1,1) is the reference point for calculating the hypervolume. For IGD, the reference set S^* is the set obtained after normalizing the non-dominated objective values of the set F.

$$f_k' = \frac{f_k - min\,(F_k)}{max\,(F_k) - min(F_k)}, k = 1,2 \tag{21}$$

The performance metrics for each algorithm are calculated using the corresponding normalized POSs. The median, average, maximum or minimum, and standard deviation of these performance metrics for case-I and case-II are shown in Table 8. The best values are marked in bold. The above comparison can also be visualized using line charts in Fig. 9, which plot the Average hypervolume, Average IGD, Average NPS, and Average time of each algorithm. The results suggest that BNSGA-II is the best performing algorithm in all performance metrics. The Pareto fronts using the six algorithms for case-I and case-II are shown in Fig. 10-11. The run of each algorithm with maximum hypervolume is used to plot these Pareto fronts. Here each Pareto optimal solution represents a trade-off squad. The $'x-$ coordinate$'$ ($'y-$ coordinate$'$) represents the normalized -net batting (bowling) performance of a squad. The extreme (toward origin) squad on the x-axis (y-axis) is best in batting (bowling) performance. The figures also show that BNSGA-II dominates the other algorithms, especially MNSGA-II, MOPSO-CD, and RNSGA-II.

TABLE 9. Friedman ranking based on hypervolume and IGD, and its test statistics.

	Case-I		Case-II	
	1/HV	IGD	1/HV	IGD
MNSGA-II	5.80	5.80	5.33	5.28
RNSGA-II	3.53	3.00	3.45	3.40
INSGA-II	2.80	3.08	1.85	1.88
BNSGA-II	**1.03**	**1.13**	**1.20**	**1.20**
NSDE	2.65	2.80	3.50	3.53
MOPSO-CD	5.20	5.20	5.68	5.73
N	40	40	40	40
Chi-square	177.357	168.471	183.729	183.343
df	5	5	5	5
p-value	8.55E-38	1.03E-37	1.96E-36	1.54E-34

This result is further investigated statistically by performing the Friedman test [23], which provides the algorithms' ranking and analyses whether the results evaluated by different algorithms demonstrate any inequality. The null hypothesis assumes that the performance of all the algorithms is equivalent or that none of the algorithms performs significantly differently. Therefore, the rejection of the null hypothesis shows a significant difference in the performance of the algorithms. IBM SPSS is used to perform this test with critical statistical significance $\alpha = 0.05$. Here, the Friedman test is performed for ranking the six algorithms (for each case separately) on the basis of their hypervolumes^{-1} and IGDs on 40 runs. The Friedman mean rank and the corresponding test statistics are shown in Table 9. As all *p-values* are less than 0.05, we reject the null hypothesis and accept that a significant difference exists in the performance of the algorithms. BNSGA-II has the highest mean rank in terms of hypervolume and IGD. NSDE has the second-highest mean rank in both metrics for case-I, and INSGA-II has the second-highest

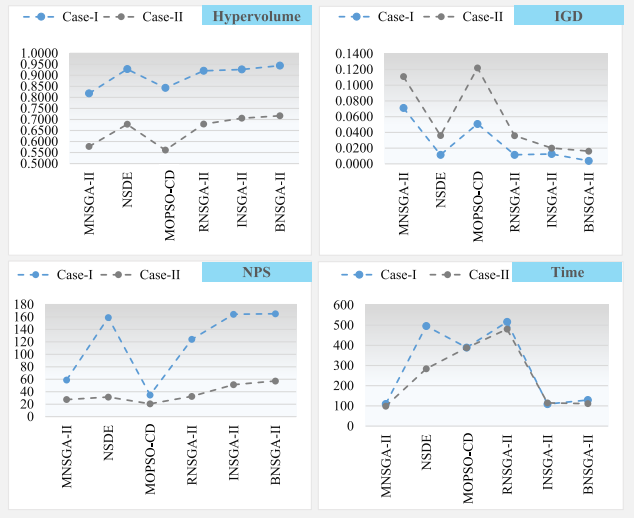


FIGURE 9. Comparison of algorithms based on hypervolume, IGD, NPS, and time.

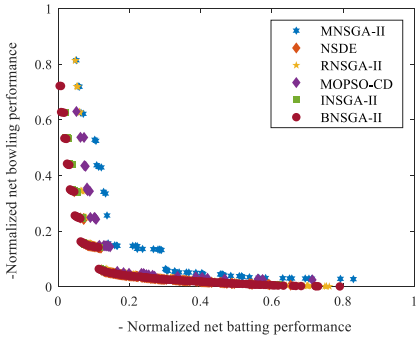


FIGURE 10. Comparison of Pareto fronts of algorithms for case-I.

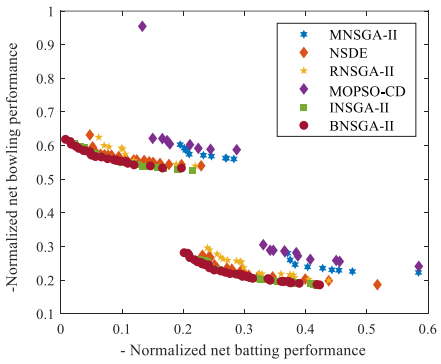


FIGURE 11. Comparison of Pareto fronts of algorithms for case-II.

mean rank for case-II. However, INSGA-II is more computationally efficient than NSDE, as can be seen in Tables 7-8. Further, MNSGA-II and MOPSO-CD have the lowest mean rank among all the compared algorithms.

2) SQUADS WITH PREFERRED PLAYERS

The franchise owners usually prefer some particular players they want in the squad regardless of the players' cost. For instance, suppose 'Rohit Sharma' (the captain of the maximum times IPL winning team) and 'Jasprit Bumrah' (one of the best death-overs bowlers) are the preference of a franchise owner for building a squad. Then, it is possible to obtain balanced trade-off squads along with these preferred players using BNSGA-II. For this purpose, we use a repair mechanism modifying Algorithm 1 such that the index of these two players has '1' in the chromosome representation (Fig.3) to maintain these two players in the squad, as shown in Table 10. In this way, BNSGA-II helps incorporate the preference of the franchise owner in squad selection which is helpful in a dynamic environment, such as an auction.

TABLE 10. Squads with preferred players.

	Squads
Squad with the highest price f(1) = -9.1839 f(2) = 972.6048 Price = ₹8175 L	Anirudha Ashok Joshi, Dale Steyn, Darshan Nalkande, David Miller, Devdutt Padikkal, Dwayne Bravo, Hardik Pandya, Jasprit Bumrah, Josh Philippe, KL Rahul, Krunal Pandya, Lungi Ngidi, M Siddharth, Monu Kumar, R Sai Kishore, Rashid Khan, Rohit Sharma, Ruturaj Gaikwad, Shane Watson, Surya Kumar Yadav, Tajinder Dhillon, Tom Banton, Tushar Deshpande
Squad with the lowest price f(1) = -9.2430 f(2) = 974.1305 Price = ₹7375 L	Anirudha Ashok Joshi, Darshan Nalkande, David Miller, Devdutt Padikkal, Dwayne Bravo, Hardik Pandya, Imran Tahir, Ishan Porel, Jasprit Bumrah, KL Rahul, Lungi Ngidi, M Siddharth, R. Sai Kishore, Rashid Khan, Rohit Sharma, Ruturaj Gaikwad, Sanjay Yadav, Shane Watson, Shubman Gill, Surya Kumar Yadav, Tajinder Dhillon, Tom Banton, Tushar Deshpande

VII. ANALYSIS AND DISCUSSION

This section analyses case-I and case-II on the basis of the trade-off squads obtained using BNSGA-II. It also analyses the trade-off squads based on cost, fielding performance, and players' performance in IPL 2020.

A. ANALYSIS OF DIFFERENT CASES

Before analyzing the trade-offs for case-I and case-II, we first obtain the trade-off squads that maximize the net batting and bowling performance and satisfy the constraints (5-9) that are based only on IPL regulations. The majority of players in almost all squads are bowlers or all-rounders, as shown in Fig. 12. Another concern is that some squads have no bowlers, which is impossible in the actual situation as an IPL squad should have the right proportion of pure batters, pure bowlers, and all-rounders.

Further, case-I bounds the sum of pure batters (bowlers) and batting all-rounders (bowling all-rounders); however, as shown in Fig. 13, in most squads, more than 60% of players are all-rounders. The reason for this is that all-rounders contribute to both the batting and bowling performance of the squad. So, the result suggests that the squads using case-I are also not balanced concerning players' expertise.

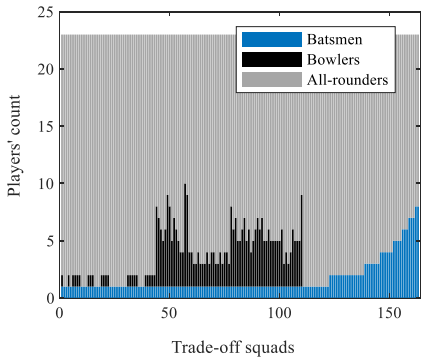


FIGURE 12. Players' count in squads formed using IPL regulations only.

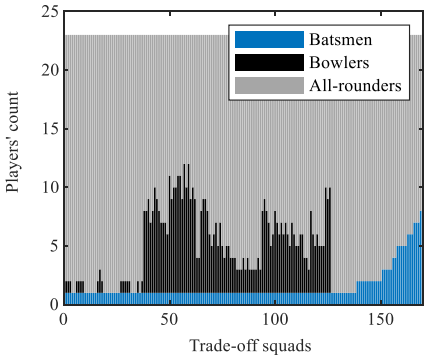


FIGURE 13. Players' count in squads for case-I.

Unlike this, a suitable proportion of pure batters, pure bowlers, and all-rounders are present in each squad for case-II, as shown in Fig.14. This result is promising since an efficient playing XI team can be made through these squads. The above analysis shows the motivation for developing a new model for IPL squad selection.

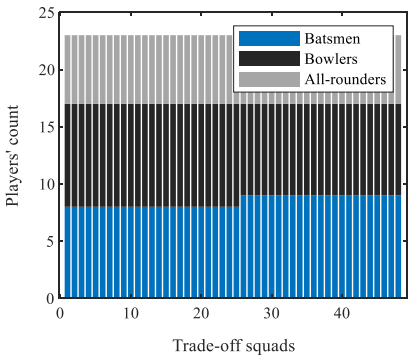


FIGURE 14. Players' count in squads for case-II.

B. TRADE-OFF SQUADS ANALYSIS

Case-II provided balanced squads; therefore, the trade-off squads obtained using BNSGA-II for this case are further analyzed based on the cost, fielding performance, and players' performance in IPL 2020.

1) SQUADS COST

The '-net batting performance versus price' and '-net bowling performance versus price' is shown in Fig.15. The figure illustrates that a squad with higher batting (bowling) performance does not need to be costlier than the squad with comparatively lower batting (bowling) performance. More clearly, two squads with similar costs may have immense differences in their batting and bowling performances. Another finding is that the squad having high net batting performance is mostly costlier than the squad with high net bowling performance, as auction prices of pure batters are higher than pure bowlers.

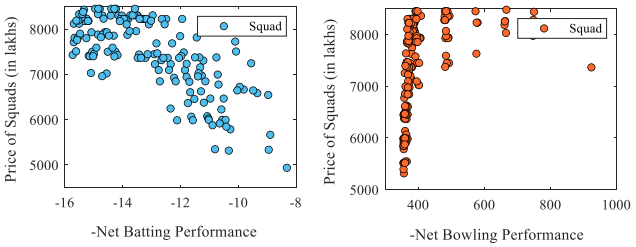


FIGURE 15. Price of squads versus batting and bowling performance of the squads.

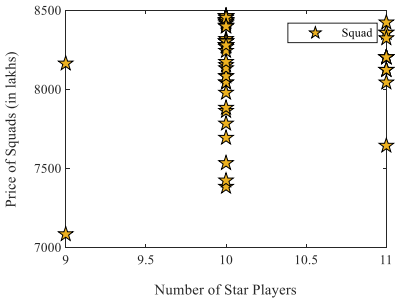


FIGURE 16. Price of squads versus the number of star players in the squads.

Fig. 16 shows the total price versus the number of star players in a squad. The presence of star players increases a squad's net batting and bowling performance. However, it is clear from the figure that a squad with more star players not necessarily be more expensive than a squad with fewer star players. It also demonstrates that many squads have the same number of star players but different total costs. The trade-off squads of the run with maximum hypervolume are used for this analysis.

TABLE 11. Wicket-keeper in trade-off squads.

Wicket-Keepers	Fielding average
AB de Villiers	0.77
KL Rahul	0.49
Rishabh Pant	0.75
Tom Banton	0.71

TABLE 12. Percentage of players' frequency in trade-offs squads.

Players	f (%)	Players	f(%)
Akash Singh	100	Sandeep Lamichhane	45
Chris Gayle	100	**Rashid Khan**	24
Darshan Nalkande	100	**Mohammad Nabi**	40
Devdutt Padikkal	100	Lalit Yadav	30
Imran Tahir	100	**Aaron Finch**	28
KL Rahul	100	Monu Kumar	27
M Siddharth	100	**Shane Watson**	27
R. Sai Kishore	100	**Glenn Maxwell**	23
Rishabh Pant	100	Anirudha Ashok Joshi	20
Virat Kohli	100	**MS Dhoni**	20
Andre Russell	98	**Abdul Samad**	15
Kieron Pollard	98	Mohsin Khan	15
Amit Mishra	97	**Shahbaz Ahmed**	15
Ruturaj Gaikwad	90	**Hardik Pandya**	12
Abhishek Sharma	88	**Rahul Tewatia**	12
Tajinder Dhillon	72	**David Miller**	8
Shubman Gill	65	**Sunil Narine**	8
Tom Banton	65	**Harpreet Brar**	7
Tushar Deshpande	65	**Kuldeep Yadav**	2
AB de Villiers	62	**Moeen Ali**	2
Lasith Malinga	62	**Nikhil Shankar Naik**	2
Pavan Deshpande	48	Suresh Raina	2

TABLE 13. Performance of trade-off squad players in IPL 2020.

Players	IPL 2020 Records
AB de Villiers	10th most runs,9th best batting average, 8th best batting strike rate,7th,8th & 9th fastest fifties, 8th most sixes, 3rd most fifties
Mohammad Nabi	2nd best bowling economy
Hardik Pandya	3rd best batting strike rate, 4th fastest fifties, 3rd most sixes
Abdul Samad	5th best batting strike rate
Kieron Pollard	6th best batting average, 3rd fastest fifty, 9th most sixes, best batting strike rate
Rashid Khan	6th most wickets, 3rd best bowling average,3rd most dot balls, best bowling economy, 6th most maiden overs, 4th highest player points
Ruturaj Gaikwad	7th best batting average
Chris Gayle	7th highest score (inning), 6th most sixes
Devdutt Padikkal	8th most runs, 5th most fours, 2nd most fifties
KL Rahul	Most runs, 5th best batting average, 2^{nd}most centuries, 7th most sixes, 3rd most fours, most fifties, 5th fastest centuries, highest player points
Rahul Tewatia	9th highest player points
Virat Kohli	9th most runs, 8th highest score (inning)
Shubman Gill	7th most fours

2) FIELDING PERFORMANCE

A player's fielding performance accounts for the number of catches, stumped out, and wickets. However, there is no data regarding the number of runs saved by a player. With this incomplete information, we cannot say about the fielding performance of a player. However, this data can estimate a wicket-keeper's performance in terms of the fielding average (22), as the higher the fielding average of a wicket-keeper, the better its fielding performance. Table 11 shows the wicket-keepers for a trade-off squad; here, AB de Villiers is the best choice for wicket-keeping.

$$Fielding\ average = \frac{Number\ of\ runout + Number\ of\ catches + Number\ of\ Stumpout}{Number\ of\ matches\ played} \tag{22}$$

3) PLAYERS' PERFORMANCE IN IPL 2020

The trade-off squads used for this analysis (BNSGA-II' run with maximum hypervolume) have total 44 players occurring in different combinations, as given in Table 12.

A player's frequency defines the number of times they occur in all the possible trade-off squads. The players with higher frequency are the key players for a squad. First, the frequency of each of the 44 players is calculated to identify the top players. Next, the frequency percentage of each player is calculated using (23).

$$f(\%) = \frac{Frequency\ of\ player}{Total\ number\ of\ trade-off\ squads} \times 100 \tag{23}$$

Out of 44 players, 10 with 100% frequency in the trade-off squads are Akash Singh, Chris Gayle, Darshan Nalkande, Devdutt Padikkal, Imran Tahir, KL Rahul, M Siddharth, R. Sai Kishore, Rishabh Pant, and Virat Kohli. 14 of them did not play in IPL 2020. The 30 players who played are marked in bold, and their performance records are obtained from *iplt20.com*. The players with one of the top ten records in IPL 2020 (in terms of most runs, most runs (over), best batting average, best batting strike rate, fastest fifties, most fifties, fastest centuries, most centuries, most fours, most sixes, highest score (innings), best bowling average, best bowling economy, best bowling strike rate, most wickets, most dot balls, most maiden overs and player points) are shown in Table 13. The remaining players are mainly in the top twenty of these records. The above analysis supports the claim that the players selected based on performance could perform well in the upcoming matches.

VIII. CONCLUSION AND FUTURE SCOPE

An efficient playing XI team for IPL implies a balanced team where a sufficient number of players with different expertise are present. The model proposed in the present study tries to form a balanced squad for IPL and has the following potential:

- It ensures the presence of a sufficient number of pure batters, pure bowlers, and all-rounders
- It evaluates the performance of a batter by estimating his batting performance through a batting performance factor which takes count of both batting average and batting strike rate and can prioritize one over the other.
- It evaluates the performance of a bowler through a combined bowling rate, which includes all three bowling statistics, i.e., bowling average, bowling strike rate, and economy rate.
- It assures that there are a sufficient number of star pure batters, star pure bowlers, and star all-rounders under the

same budget for better performance and to meet franchise business prospects.
- The performance of the players is assessed not only through the IPL data but also through the data of other tournaments played in the T20 format.

This study identifies the squad selection problem as a knapsack problem due to the nature of the decision for a player selection (yes or no) and the objective of maximizing squad performance under a fixed budget. On this basis, two combinatorial optimization algorithms BNSGA-II and INSGA-II, are developed to obtain its trade-off solutions. The proposed algorithms are validated by comparing them with the existing optimization algorithm for IPL team selection and with the three other prominent algorithms, RNSGA-II, NSDE, and MOPSO-CD, using the data of players auctioned in IPL 2020. Overall, this comparison demonstrates BNSGA-II as the best-performed algorithm in terms of hypervolume and IGD and reports similar performances of NSDE and INSGA-II, though INSGA-II is computationally efficient than NSDE

Further, it is observed that many of the players selected through the proposed model played well in IPL2020. This model can be an additional tool for the selectors and the franchise owners to make a more concrete, unbiased, and systematic decision for the selection of the squad. Other than cricket, there can be other sports also where such a model can be implemented. At a broader level, this being a selection problem can be implemented under any scenario where a limited selection is to be done out of the given sample.

The present study tries to develop a model that is appropriate for selecting players an IPL squad. However, there are some limitations that may be addressed in a future study:

- In this study, the players' past information has been used to obtain the trade-off squads. Therefore, some amount of risk is involved in making such a decision because a player's performance may not be the same in the upcoming matches. So whether more weightage is to be given to the in-form player in comparison to the out-of-form player may also be considered while designing the model.
- The most prominent factors, like batting average, bowling average etc., are considered in this study for designing a well-balanced team. However, there are other important features also, like the condition of the pitch, home ground etc., that may also be considered while designing the model for a well-balanced team.
- In cricket, there are several forms of bowling and batting. Like in the case of bowling, there are fast bowlers, medium-pace bowlers, and spinners, and for batting, there are openers and middle-order batters. Further studies may be done to include all these types while developing a model.

In addition, it will be interesting to design a graphical user interphase for the selection of a balanced squad.

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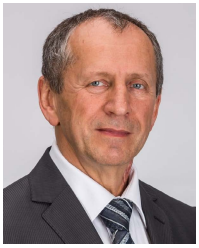
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